

STANDARD OPERATING PROCEDURE (SOP)

Ubuntu Linux Virtual Machine Setup Using VirtualBox

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Operating System: Ubuntu Linux

Virtualization Platform: Oracle VirtualBox

1. Purpose

This Standard Operating Procedure (SOP) documents the process for installing and configuring an Ubuntu Linux virtual machine using Oracle VirtualBox.

The purpose of this procedure is to provide a repeatable process for:

- Creating an Ubuntu virtual machine
 - Configuring virtual hardware
 - Installing Ubuntu Linux
 - Configuring the operating system
 - Installing required updates and software
 - Verifying the system is functioning correctly
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2. Scope

This SOP applies to the setup and initial configuration of an Ubuntu Linux virtual machine in Oracle VirtualBox.

This procedure is intended for:

- Linux system administration practice
- Virtualization practice
- Cybersecurity labs
- IT administration training

- Development and testing environments
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3. Prerequisites

Before beginning this procedure, verify that the following requirements are available.

Hardware Requirements

- Windows/macOS/Linux host computer
- Minimum 8 GB available RAM
- Minimum 30 GB available storage

Software Requirements

- Oracle VirtualBox installed
- Ubuntu ISO image downloaded
- Internet connection available

Required Files

Ubuntu ISO:

[Ubuntu ISO Download Link](#)

VirtualBox Install:

[VirtualBox Install](#)

4. Procedure

Step 1 — Install Ubuntu ISO

Objective

Install the newest version of the Ubuntu ISO for our machine.

Procedure

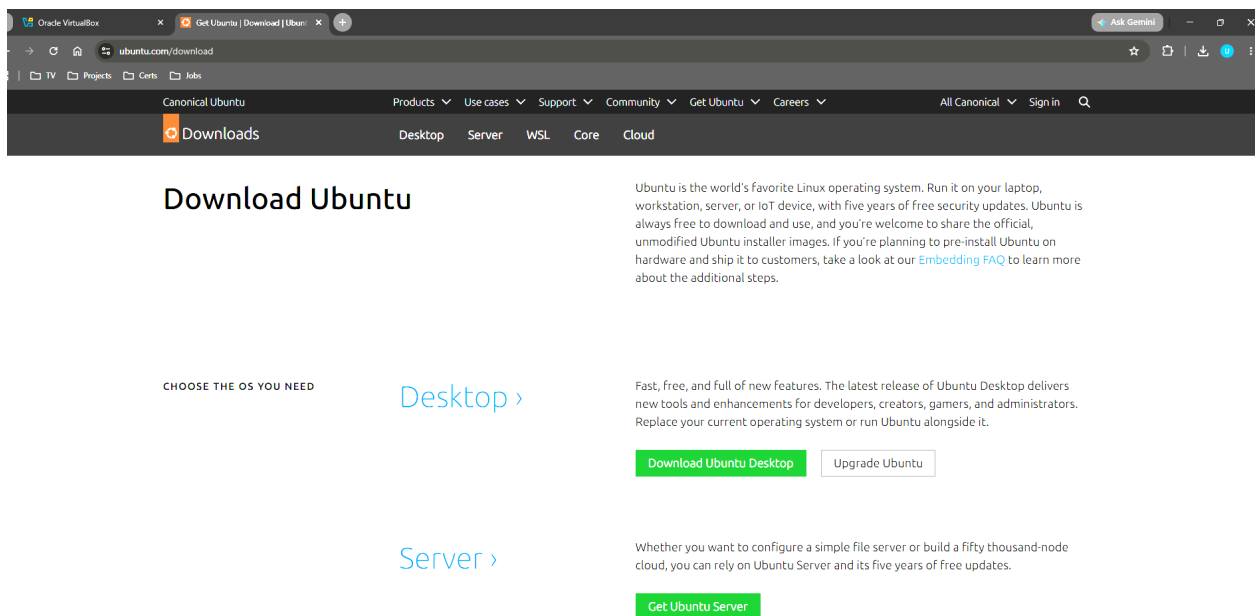
1. Visit <https://ubuntu.com/download>

For this setup, choose of these:

- **Ubuntu Desktop** if you want a GUI
 - **Ubuntu Server** if you want only command line
2. Install whichever OS version of Ubuntu you want to use (I used Desktop in this SOP)
 3. Save the ISO file in a place where you can easily access or remember where it was installed.

Screenshot

Figure 1 — Screenshot of the ubuntu ISO install page.



Notes:

Desktop is better if you are new to Linux and want a Graphical User Interface (GUI)

Server is better if you are familiar with Linux as it provides only the command line

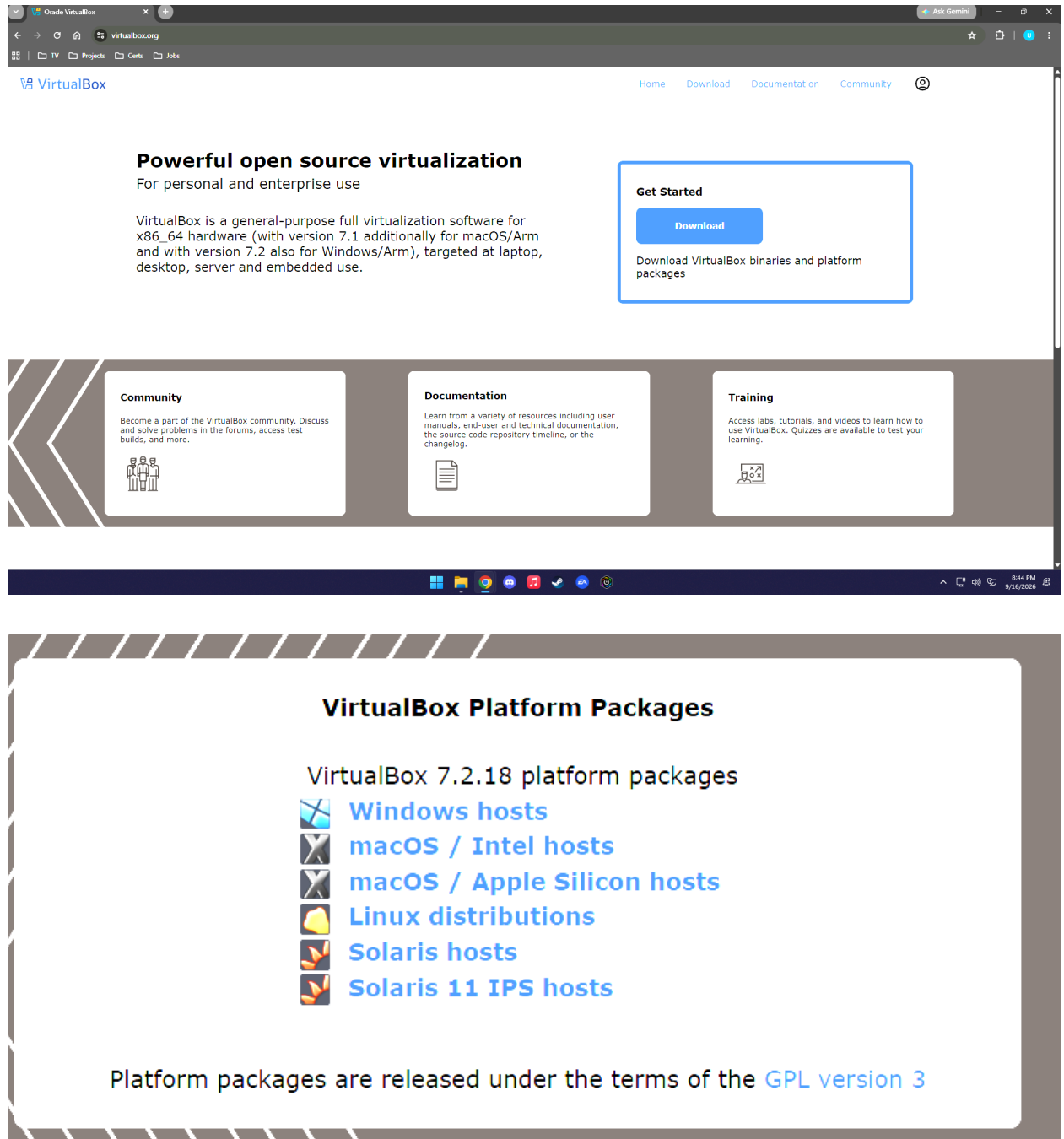
Step 2 — Installing VirtualBox

Objective

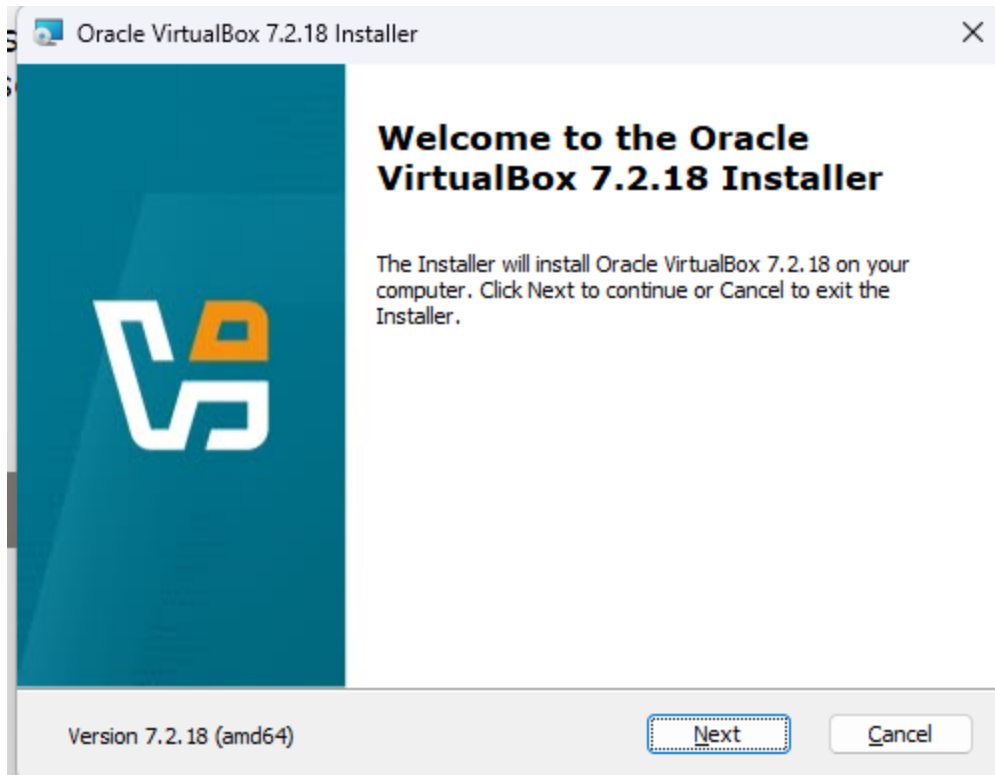
Install and configure the VirtualBox Application.

Procedure

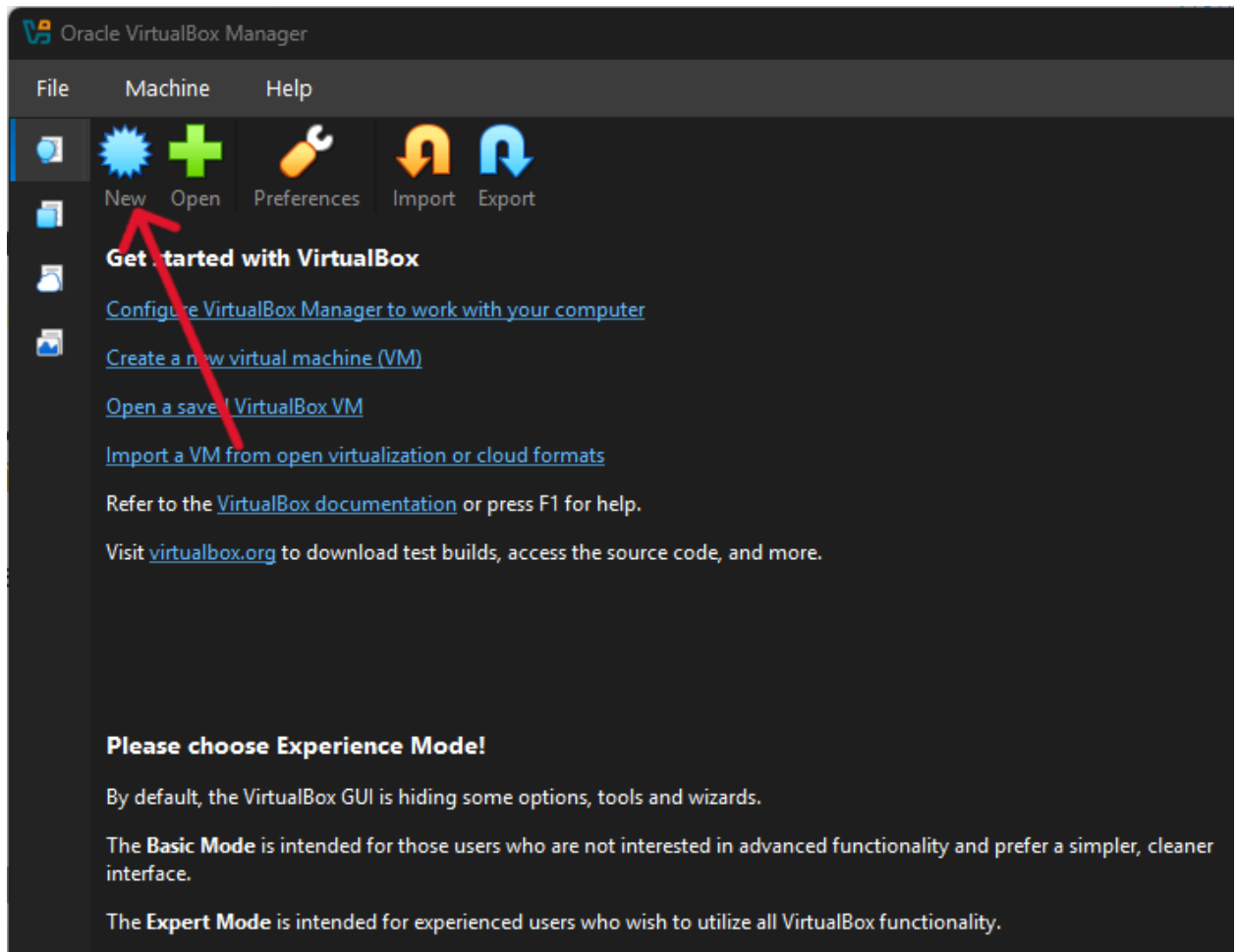
1. Visit <https://www.virtualbox.org/> to install the VirtualBox software (choose your current OS).



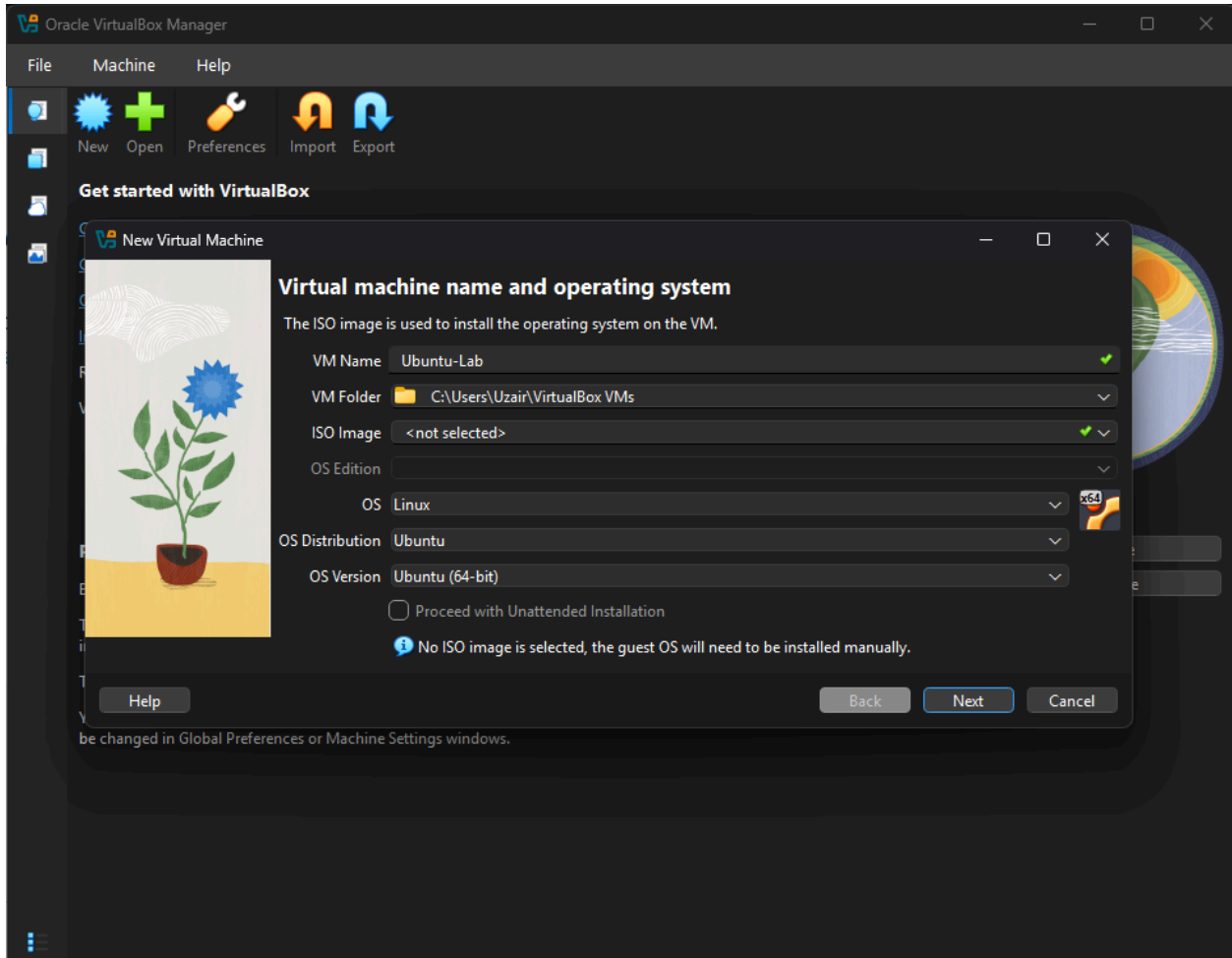
2. After download is complete, open the download file and run through the setup process.

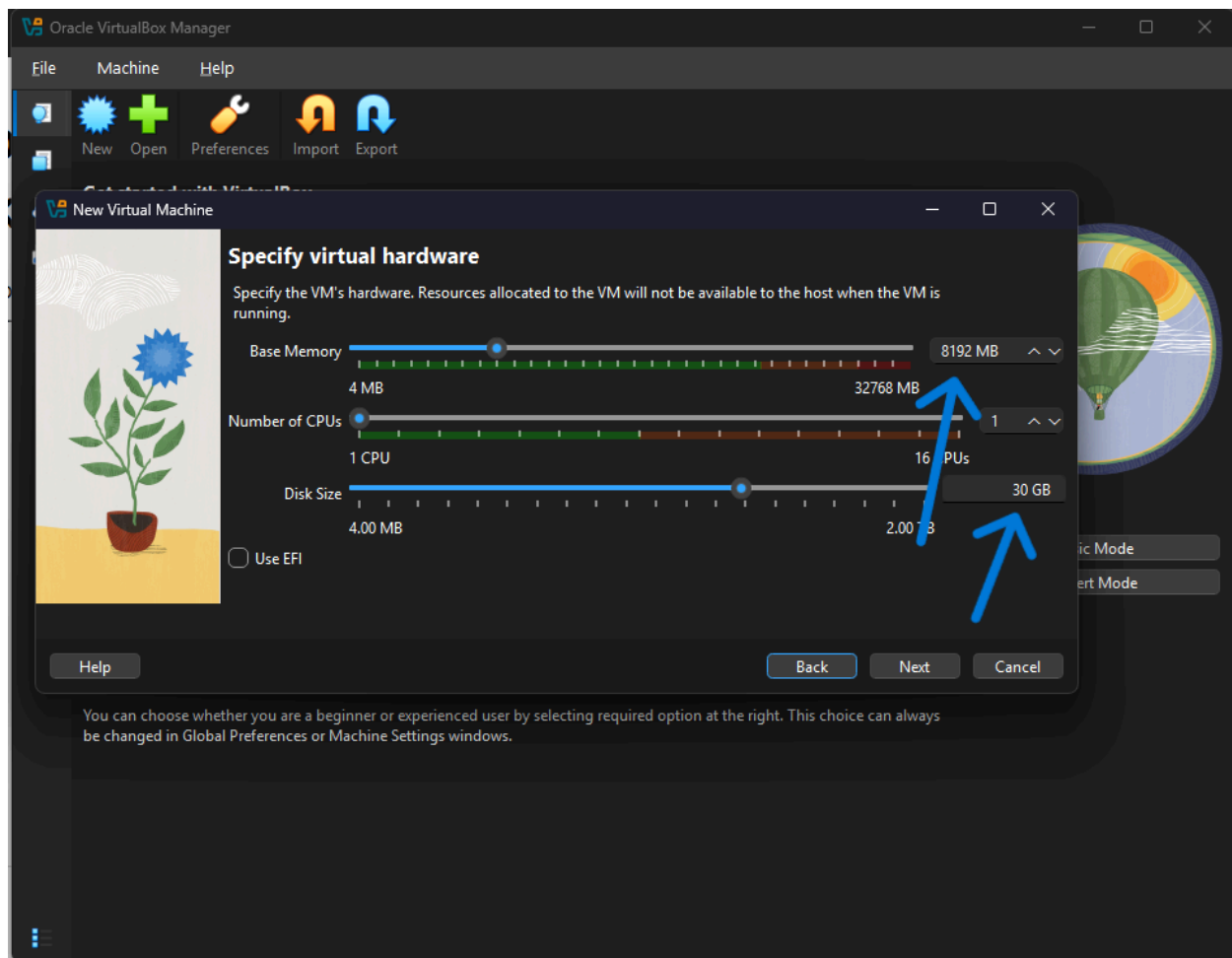


3. Once VirtualBox is installed and gets to the home screen, click new in the top left to get the configuration for our virtual machine started.



4. You will be greeted with a new page. This is where you can name your virtual machine. I called mine "Ubuntu-Lab." You can also attach the ISO file and configure the RAM and memory.
 - **NOTE:** If your system has 8GB RAM, you want to configure the system to 4096 MB minimum. If your system has more than 8GB RAM, then set the memory to 8192 MB.





Notes:

If you want, you can increase the memory and disk size. It is better to have at least 30GB of disk size as we will be installing tools and packages

Step 3 — Start the Virtual Machine

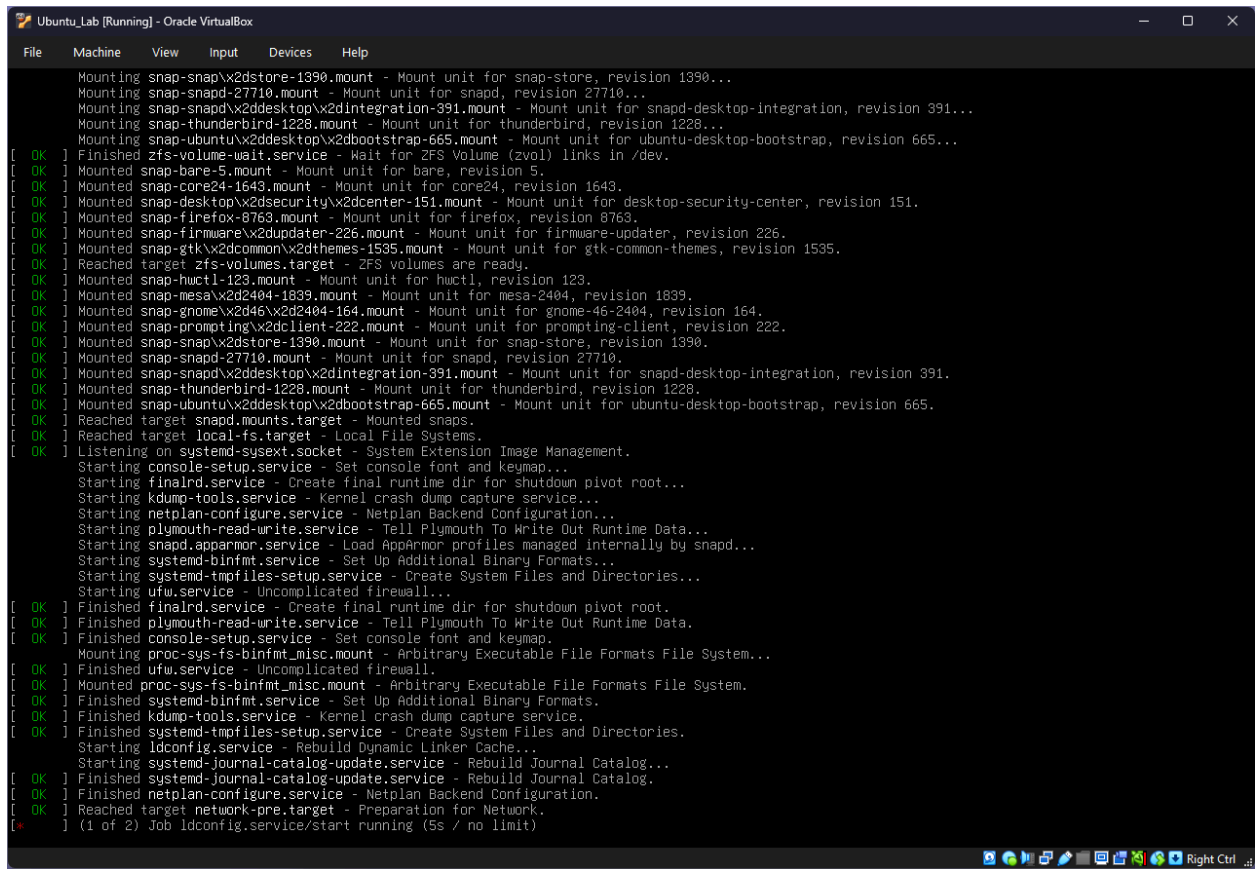
Objective

Start the newly created VM and begin the Ubuntu installation process.

Procedure

1. After everything is set up and configured, go ahead and run the virtual machine.

- The VM might take a minute to start, but once it does, install Ubuntu when the prompt is given and you will see a terminal pop up with the Ubuntu being installed



```
Mounting snap-snap\x2dstore-1390.mount - Mount unit for snap-store, revision 1390...
Mounting snap-snapd-27710.mount - Mount unit for snapd, revision 27710...
Mounting snap-snapd\x2ddesktop\x2dintegration-391.mount - Mount unit for snapd-desktop-integration, revision 391...
Mounting snap-thunderbird-1228.mount - Mount unit for thunderbird, revision 1228...
Mounting snap-ubuntu\x2ddesktop\x2dbootstrap-665.mount - Mount unit for ubuntu-desktop-bootstrap, revision 665...
[ OK ] Finished zfs-volume-wait.service - Wait for ZFS Volume (zvol) links in /dev.
[ OK ] Mounted snap-bare-5.mount - Mount unit for bare, revision 5.
[ OK ] Mounted snap-core24-1643.mount - Mount unit for core24, revision 1643.
[ OK ] Mounted snap-desktop\x2dsecurity\x2dcenter-151.mount - Mount unit for desktop-security-center, revision 151.
[ OK ] Mounted snap-firefox-8763.mount - Mount unit for firefox, revision 8763.
[ OK ] Mounted snap-firmware\x2dupdater-226.mount - Mount unit for firmware-updater, revision 226.
[ OK ] Mounted snap-gtk\x2dcommon\x2dthemes-1535.mount - Mount unit for gtk-common-themes, revision 1535.
[ OK ] Reached target zfs-volumes.target - ZFS volumes are ready.
[ OK ] Mounted snap-hucl1-123.mount - Mount unit for hucl1, revision 123.
[ OK ] Mounted snap-mesa\x2d2404-1839.mount - Mount unit for mesa-2404, revision 1839.
[ OK ] Mounted snap-gnome\x2d46\x2d2404-164.mount - Mount unit for gnome-46-2404, revision 164.
[ OK ] Mounted snap-prompting\x2dclient-222.mount - Mount unit for prompting-client, revision 222.
[ OK ] Mounted snap-snap\x2dstore-1390.mount - Mount unit for snap-store, revision 1390.
[ OK ] Mounted snap-snapd-27710.mount - Mount unit for snapd, revision 27710.
[ OK ] Mounted snap-snapd\x2ddesktop\x2dintegration-391.mount - Mount unit for snapd-desktop-integration, revision 391.
[ OK ] Mounted snap-thunderbird-1228.mount - Mount unit for thunderbird, revision 1228.
[ OK ] Mounted snap-ubuntu\x2ddesktop\x2dbootstrap-665.mount - Mount unit for ubuntu-desktop-bootstrap, revision 665.
[ OK ] Reached target snapd.mounts.target - Mounted snaps.
[ OK ] Reached target local-fs.target - Local File Systems.
[ OK ] Listening on systemd-sysext.socket - System Extension Image Management.
Starting console-setup.service - Set console font and keymap...
Starting finalrd.service - Create final runtime dir for shutdown pivot root...
Starting kdump-tools.service - Kernel crash dump capture service...
Starting netplan-config.service - Netplan Backend Configuration...
Starting plymouth-read-write.service - Tell Plymouth To Write Out Runtime Data...
Starting snapd.apparmor.service - Load AppArmor profiles managed internally by snapd...
Starting systemd-binfmt.service - Set Up Additional Binary Formats...
Starting systemd-tmpfiles-setup.service - Create System Files and Directories...
Starting ufw.service - Uncomplicated firewall...
[ OK ] Finished finalrd.service - Create final runtime dir for shutdown pivot root.
[ OK ] Finished plymouth-read-write.service - Tell Plymouth To Write Out Runtime Data.
[ OK ] Finished console-setup.service - Set console font and keymap.
Mounting proc-sys-fs-binfmt_misc.mount - Arbitrary Executable File Formats File System...
[ OK ] Finished ufw.service - Uncomplicated firewall.
[ OK ] Mounted proc-sys-fs-binfmt_misc.mount - Arbitrary Executable File Formats File System.
[ OK ] Finished systemd-binfmt.service - Set Up Additional Binary Formats.
[ OK ] Finished kdump-tools.service - Kernel crash dump capture service.
[ OK ] Finished systemd-tmpfiles-setup.service - Create System Files and Directories.
Starting ldconfig.service - Rebuild Dynamic Linker Cache...
Starting systemd-journal-catalog-update.service - Rebuild Journal Catalog...
[ OK ] Finished systemd-journal-catalog-update.service - Rebuild Journal Catalog.
[ OK ] Finished netplan-config.service - Netplan Backend Configuration.
Reached target network-pre.target - Preparation for Network.
(1 of 2) Job ldconfig.service/start running (5s / no limit)
```

- After everything is installed, you will be brought to the home screen of your new UbuntuOS.

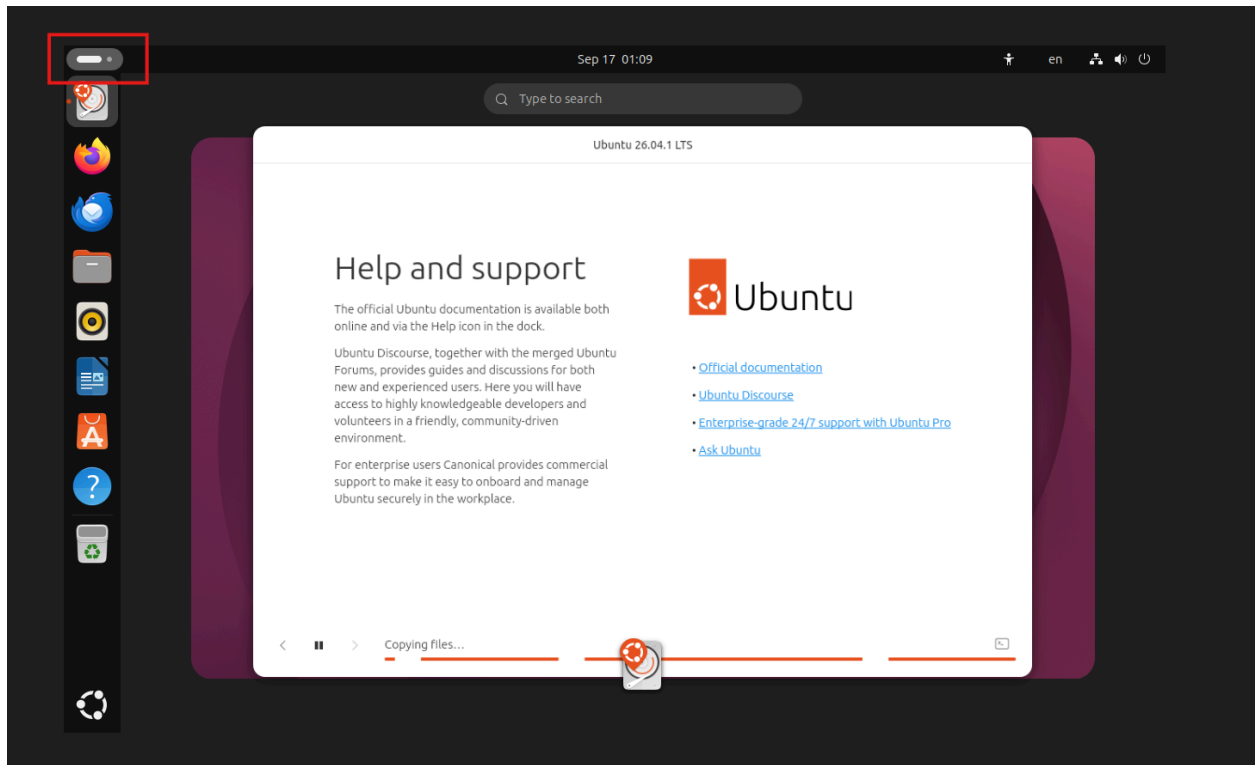
Step 4 — Finding and Opening Terminal

Objective

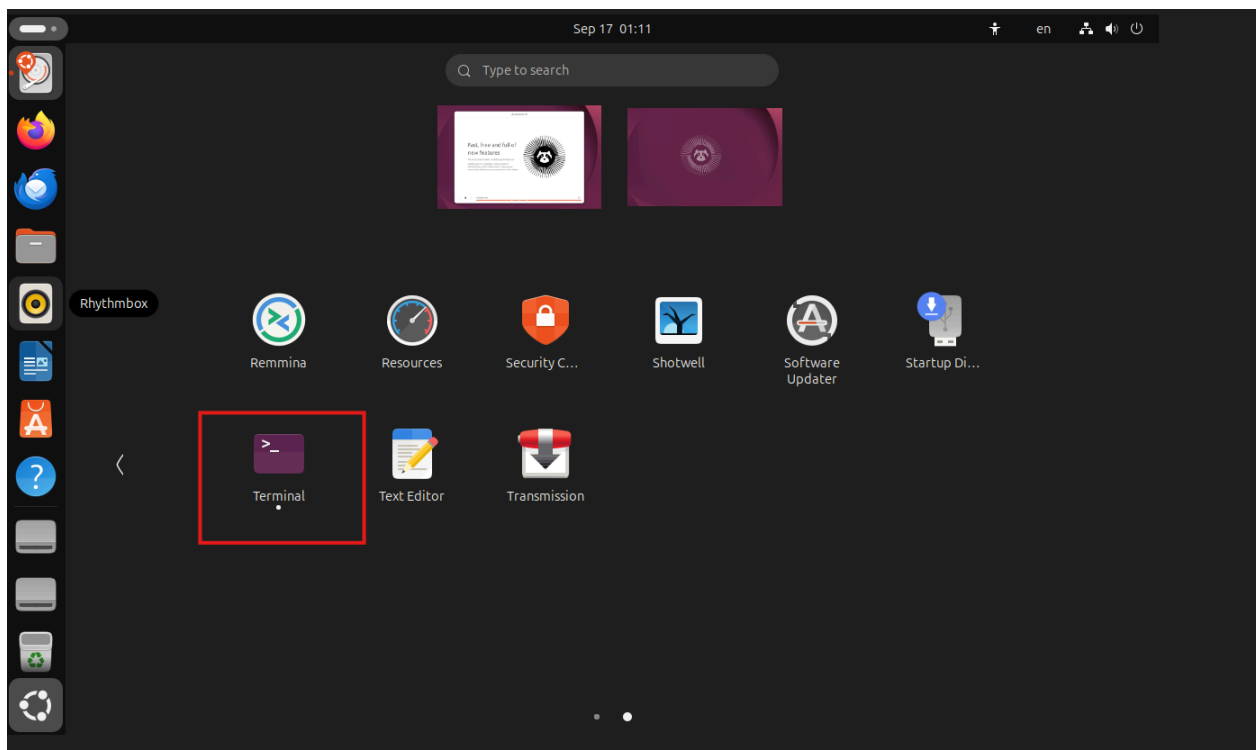
Finding and getting into Terminal.

Procedure

- There are going to be two dots on the top left of your Ubuntu home page. Click those dots to open up a vertical taskbar.



2. After the taskbar is open, you will see a Ubuntu logo at the bottom left. Click that to open the complete list of applications.
3. The terminal should be listed somewhere between all the applications



4. Double click on the terminal to open it up.

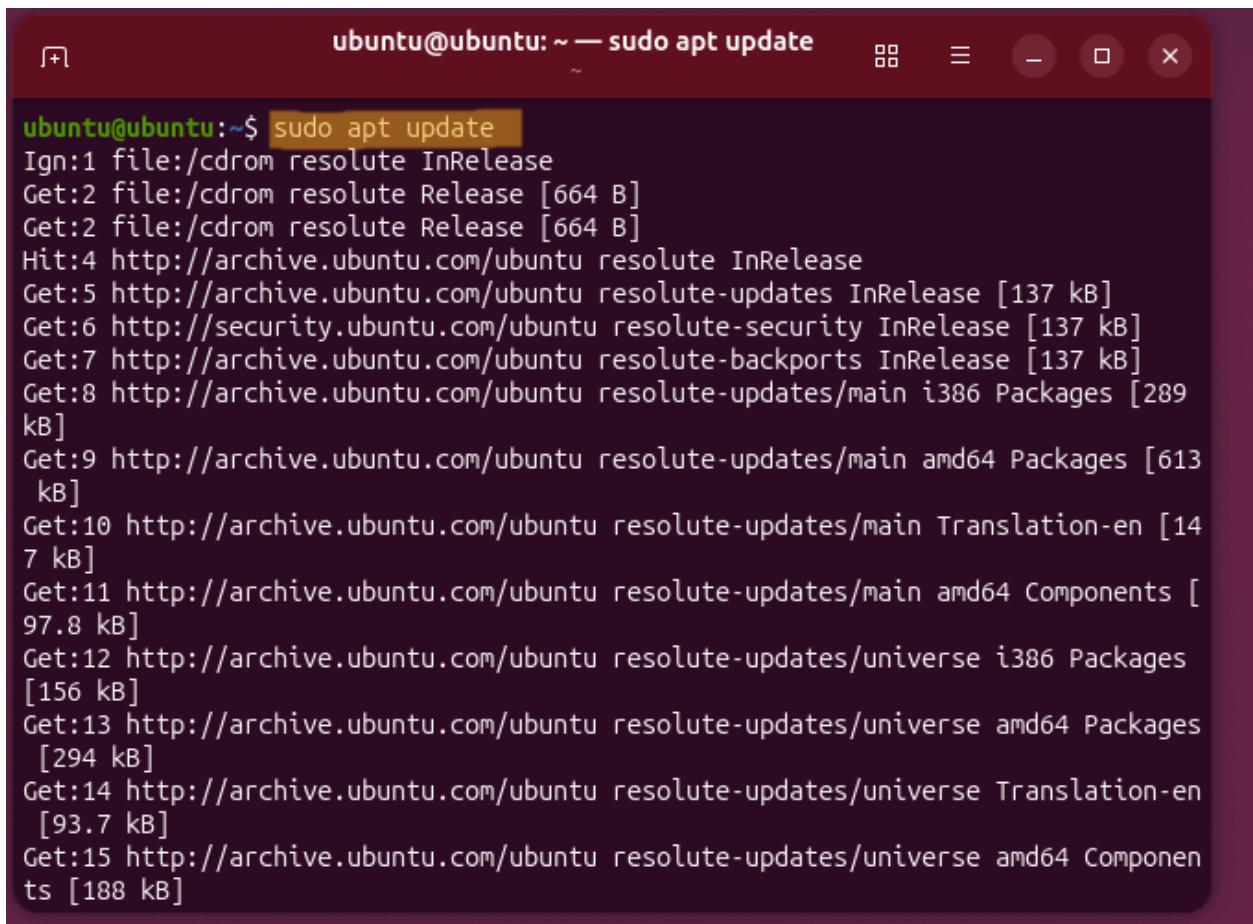
Step 5 — Running Terminal Commands and What They Mean

Objective

Run the basic and important commands in the Terminal, as well as understanding what each command does.

Procedure

1. `sudo apt update`
 - a. Checks for the latest package list

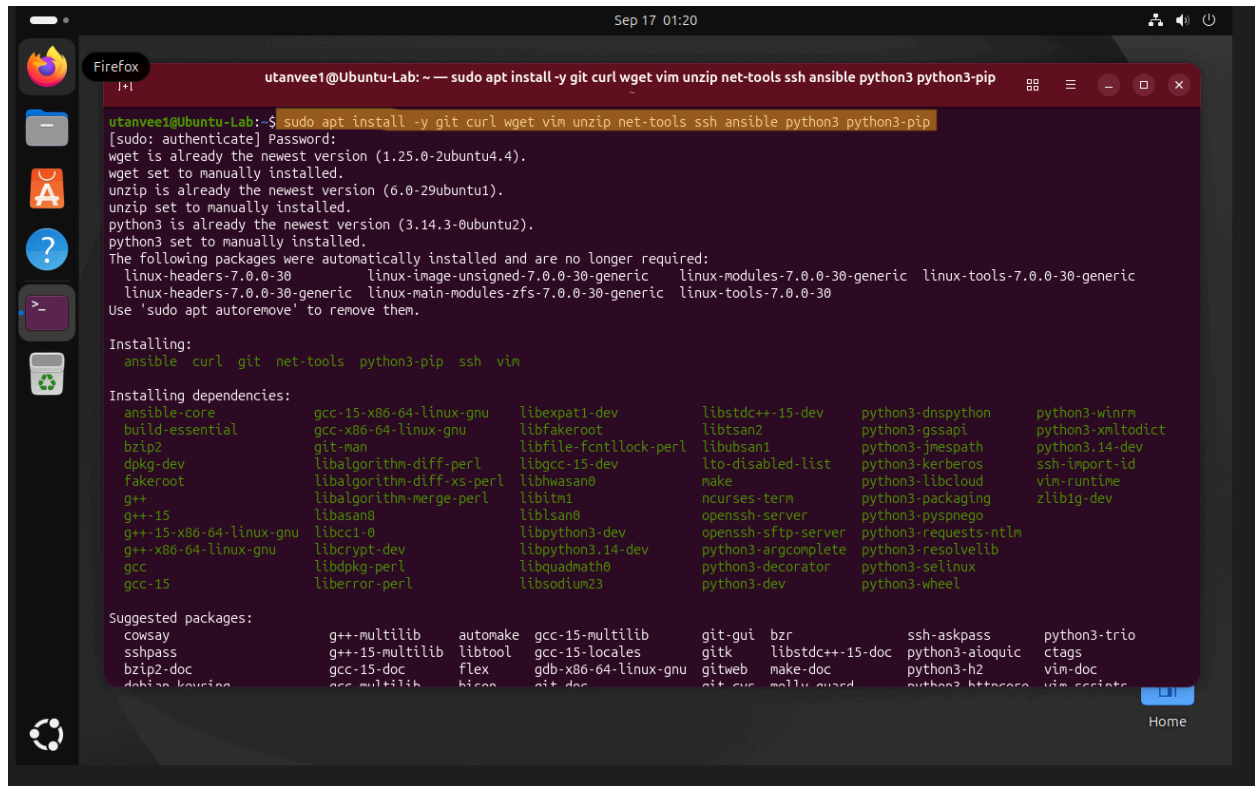


```
ubuntu@ubuntu: ~ — sudo apt update
ubuntu@ubuntu:~$ sudo apt update
Ign:1 file:/cdrom/resolute InRelease
Get:2 file:/cdrom/resolute Release [664 B]
Get:2 file:/cdrom/resolute Release [664 B]
Hit:4 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main i386 Packages [289 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [613 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [147 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/universe i386 Packages [156 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [294 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [93.7 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [188 kB]
```

2. `sudo apt upgrade -y`
 - a. Installs updates automatically

```
ubuntu@ubuntu: ~ -- sudo apt upgrade -y
155 packages can be upgraded. Run 'apt list --upgradable' to see them.
Notice: Skipping acquire of configured file 'main/binary-i386/Packages' as repository 'file:/cdrom resolute InRelease' doesn't support architecture 'i386'
Notice: Skipping acquire of configured file 'restricted/binary-i386/Packages' as repository 'file:/cdrom resolute InRelease' doesn't support architecture 'i386'
ubuntu@ubuntu:~$ sudo apt upgrade -y
Get another security update through Ubuntu Pro with 'esm-apps' enabled:
restic
Learn more about Ubuntu Pro at https://ubuntu.com/pro
Upgrading:
bind9-dnswtills      libaudit-common      libpam0g              perl
bind9-host           libaudit1             libperl5.40           perl-base
bind9-libs            libblkid1             libpolkit-agent-1-0   perl-modules-5.40
bluez                libbluetooth3         libpolkit-gobject-1-0 pkexec
bluez-cups            libbz2-1.0            libpython3.14         polkitd
bluez-obexd           libc-bin              libpython3.14-minimal python3-apt-common
bpftool              libc-dev-bin          libpython3.14-stdlib  python3-apt
bsdextrautils         libc-gconv-modules-extra libsmartcols1         python3-cryptography
bsdutils              libc6                  libsqlite3-0          python3-netplan
bzip2                 libc6-dbg              libssh-4              python3-pil
cpio                  libc6-dev              libssh2-1t64          python3-sss
diffutils             libcurl3t64-gnutls    libssl3t64            python3.14
dirmngr              libcurl4t64            libsss-certmap0       python3.14-gdbm
dmidecode             libevent-core-2.1-7t64 libsss-idmap0         python3.14-minimal
dracut-install        libevent-extra-2.1-7t64 libsss-nss-idmap0     rftkill
eject                 libfdisk1              libudisks2-0          spice-vdagent
fdisk                 libflashrom1           libuuid1              sssd
firefox               libfreerdp-client3-3   libutil-linux          sssd-ad
git-2.43.0+really2.43.1 libfsverity1            libusbkit-2t64-1-0    sssd-ad-common
```

3. `sudo apt install -y git curl wget vim unzip net-tools ssh ansible python3 python3-pip`
 - a. `git` = upload projects to GitHub
 - b. `curl` / `wget` = download files
 - c. `vim` = edit text files
 - d. `unzip` = extract files
 - e. `net-tools` = networking commands
 - f. `ssh` = remote login
 - g. `ansible` = automation
 - h. `python3` & `pip` = scripting
 - i. `pip` = allows the install of Python packages



NOTE: This step will ask for your password. Enter the password you originally used when setting up your VM to continue installing the tools

Step 6 — Enabling SSH

Objective

Check to see if SSH is installed and active. SSH is important as it is the most common Linux admin skill and is important for remote work in Linux.

Procedure

1. Open the Terminal.
2. `ssh -V`
 - a. Checks to see if SSH is installed
 - i. If not installed, run: `sudo apt install -y openssh-server`
3. `sudo systemctl status ssh`
 - a. Checks the SSH status
4. If the status is showing disabled, run:
 - a. `sudo systemctl enable ssh`

b. `sudo systemctl start ssh`

5. Revision History

Version	Date	Author	Description
1.0	09.16.26	Uzair Tanveer	Initial SOP created

Appendix A — Commands Used

Document all Linux commands used throughout the setup.

`sudo apt update` — checks for latest packages list

`sudo apt upgrade -y` — installs updates automatically

`sudo apt install -y git curl wget vim unzip net-tools ssh ansible python3 python3-pip` — installs common tools that would be useful for projects

`ssh -V` — checks if SSH is installed

`sudo apt install -y openssh-server` — installs SSH

`sudo systemctl status ssh` — checks status of SSH

`sudo systemctl enable ssh` — enables SSH

`sudo systemctl start ssh` — starts SSH
